VOIDSCOPE · CONCEPT OPTIONS

VoidScope concept options

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First issue (kit 1.9.1 upgrade): the current design is option A; options B (reel-fed push rod) and C (tripod and winch sonde) added with a different working principle each, all three modelled at D3, scored against VDS-URS-001

VoidScope concept options

Option A, the rigid sectioned rod with a steerable tip (the design of record), scores 165 against 135 for option B and 128 for option C, 22 per cent clear of the runner-up; it is recommended, and the choice is Proposed, awaiting Amish (VDS-DDR-005). Method: STANDARDS section 21.

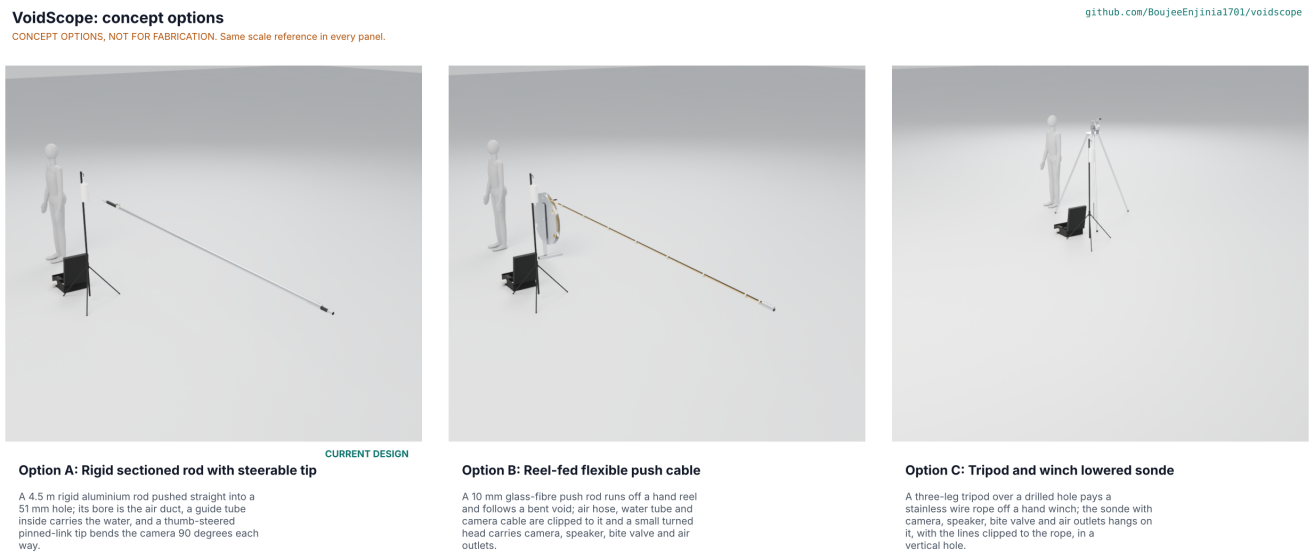


Figure 1. The three options at one scale. The picture is rendered from the D3 models in `cad/src/model.py` *by* `.kit/options.py`*; the coordinator renders it on Amish's Mac, so it is absent until then.*

> **Safety:** All three options are paper concepts for rescue equipment that may supply water and air to a trapped person. None is safe to use as drawn. VoidScope is an open engineering reference, not certified rescue equipment and not a medical device.

The options

Option A: rigid sectioned rod with a steerable tip (design of record)

Working principle: push a rigid rod straight through the hole and look round corners with a steerable camera tip. A 4.5 m rod of four 38.1 mm aluminium sections joined by snap buttons; its bore is the air duct and a guide tube inside carries the single-use water tube to a bite valve on the nose; a turned 48 mm head, a pinned-link tip bent by two wires from a thumb lever. First-order numbers: 4.13 m working length (tip rested on debris beyond about 2 m), 48 mm head, rod buckles at 940 N, 2.2 min setup, USD 1,780 (VDS-CAL-001). Key risks: the tip heat (fixed, STM S14), the drip margin (fixed, S7), the wire breaking load and the bite valve, all to confirm when parts are bought. Needs from a co-design partner: a medical lead to review the water and air lines, and training rubble.

Option B: reel-fed flexible push cable

Working principle: feed a flexible rod off a reel and let it follow the void. A 10 mm glass-fibre rod, 5.2 m, wound on a 640 mm hub between two 800 mm aluminium flanges; the air hose, water tube and camera cable are clipped beside the rod every 650 mm; a turned 40 mm head carries the camera, speaker, bite valve and six air outlets as in A, without a steerable tip. First-order numbers: strain on the reel 1.5 per cent (STM SB2, plausible); the rod droops

93 mm at 1 m in free air (STM SB1), so it reaches 4 m only where the void floor carries it. Key risks: reach in free air, a loaded rod springing back, lines outside the rod catching on rebar, a heavy reel to carry (the modelled kit weighs 26.6 kg against 11.8 kg for A). Needs: trials in a floor-supported void to learn how far it really goes.

Option C: tripod and winch lowered sonde

Working principle: lower a camera sonde on a rope down a vertical hole, as a borehole camera is. Three 32 x 2 mm aluminium legs pinned to a head plate over the hole; a hand winch (drum, crank, ratchet and pawl) pays a 4 mm stainless rope; a 44 x 360 mm sonde hangs on an eye bolt, camera down, with the lines clipped to the rope. First-order numbers: depth 4 m; each leg takes 4.5 kN before it buckles against 150 N needed (STM SC1). Key risks: it works only down a vertical hole, the sonde spins and swings on its rope, a hanging load above a survivor. Needs: a vertical drilled hole and a medical lead's view on offering water to a swinging valve.

First-order numbers

Table 1. Options compared. Parts counts, masses and step files are from `cad/fidelity.json` *(kit 1.9, D3; mass includes the shared surface unit and stand).*

Quantity	Option A	Option B	Option C
Estimated parts cost (USD)	1,780 (bom/bom.csv)	999 shared with A, plus probe parts to quote	999 shared with A, plus probe parts to quote
Mass (kg), modelled	11.8	26.6	15.5
Output or throughput	4.13 m horizontal reach, tip steers 90 degrees each way	1 m free reach; up to 5.2 m of rod where the floor carries it	4 m depth, vertical hole only
Energy or effort	One 12.8 V 6 Ah pack, 6.2 h video; hand push	Same unit; hand push and reel	Same unit; hand winch
Parts count (D3 model)	143	125	143
STEM check	22 rows, 11 within, 11 plausible; S14 tip heat and S7 capillary margin fixed	SB1 restated (free reach capped at 1 m); SB2 plausible	SC1 within; SC2 plausible

Scoring

Table 2. Weighted scoring (weights: High UR 3, Medium 2, Low 1; scores 1 to 5; the last four rows are the STANDARDS section 21 criteria).

Criterion	Source	Weight	A	B	C	Reason
Fits a 50 mm hole	UR-001	3	4	5	4	A head 48 mm leaves 1.5 mm a side; B head 40 mm and a 10 mm rod; C sonde 44 mm
Reaches 4 m along a straight path	UR-002	3	5	2	3	A 4.13 m rigid; B 1 m free, floor-supported only (STM SB1); C 4 m deep but vertical only
Sees in darkness and dust	UR-003	3	4	4	4	Same bought camera in all three
Survives rubble handling	UR-004	3	4	3	3	A aluminium rod and housing; B glass-fibre rod splinters and abrades; C rope abrades on the hole edge
Water cannot harm the survivor	UR-005	3	4	3	2	Same capillary and bite valve; A can steer the tip aside; B has no steering; C sonde swings and spins on its rope, so the valve is hard to offer
Air clean and gentle	UR-006	3	5	3	3	A uses the 35 mm bore as the duct (0.38 kPa); B and C use a 10 to 12 mm hose clipped alongside
Works a whole shift	UR-007	3	4	4	4	Same surface unit and packs in all three
Looks round a corner	UR-011	3	4	3	1	A steers 90 degrees each way; B follows a bend with the rod but cannot aim; C has no steering
Affordable for a municipal budget	UR-008	3	4	3	4	A USD 1,780 (BOM); B and C share USD 999 of A's lines; their probe parts are not priced (to quote); B adds a reel and welded upright, C a winch and tripod
Quick to set up and carry	UR-009	2	3	2	3	A 2.2 min; B reel adds mass (kit 20.2 kg); C tripod set over the hole
Water path stays clean	UR-010	2	4	3	3	A water tube runs inside a guide tube; B and C clip it outside, where rubble touches it
Talks with the survivor	UR-012	2	4	4	4	Same tip speaker in all three
Nothing hot or hazardous at the tip	UR-013	2	4	4	4	Same camera and the same black anodised finish (STM S14)
Buildable with local skills and tools	STANDARDS 21	2	3	3	3	A needs outside lathe time; B needs welding and a reel; C needs a built winch with pawl
Safety (structure, hanging loads, stops)	STANDARDS 21	3	4	3	2	C hangs a weight and rope over a possible survivor; B's rod can spring back; A has the blocked-tip stop and is steerable out
Maintenance	STANDARDS 21	1	3	4	4	B and C have fewer, simpler parts to replace
Weighted total (of 205)			**165**	**135**	**128**	A leads B by 22 per cent and C by 29 per cent

Recommendation

Recommended: option A. It is the only option that reaches 4 m horizontally, aims the camera and uses the rod bore as a clean air duct, and it scores 22 per cent above B and 29 per cent above C, outside the 10 per cent band, so one option goes forward as a proposal. This is the design of record and nothing is redesigned. Recorded in VDS-DEC-001 as "Proposed, awaiting Amish" and argued in VDS-DDR-005; `project.yaml` says `concept_chosen: A (proposed)`.

Best ideas to graft onto A from the others: from B, the reel concept for a long water and camera umbilical (the 7 m cable and 8 m tube today lie loose on the ground); from C, a tripod head plate as an optional guide for pushing the rod into a vertical hole, which makes a vertical entry steadier. Both are suggestions, not changes.